

Juan C. Yamin

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2026–27 Economics Job Market Candidate

Fields: Econometrics, Development Economics

Research interests: Statistical decision theory, empirical Bayes, experimental design, causal inference

DISSERTATION COMMITTEE AND REFERENCES

Toru Kitagawa

Professor of Economics
Brown University
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Soonwoo Kwon

Assistant Professor of Economics
Brown University
soonwoo_kwon@brown.edu

Jonathan Roth

Professor of Economics
Brown University
jonathan_roth@brown.edu

EDUCATION

Brown University, Providence, RI

Ph.D. in Economics

Expected May 2027

M.A. in Economics

May 2022

Universidad de los Andes, Bogotá, Colombia

M.Sc. in Economics

March 2019

B.Sc. in Economics

October 2017

JOB MARKET PAPER

Poverty Targeting with Imperfect Information

Targeted antipoverty programs in developing countries must rely on estimated rather than observed income, which leads to substantial targeting errors. I formulate the allocation of a fixed transfer budget based on those estimates as a statistical decision problem and show that the standard plug-in rule is inadmissible. I then develop a nonparametric empirical Bayes rule whose Bayes regret is bounded by the estimation error of the posterior mean poverty gaps and therefore inherits their rate of convergence. In simulations using household surveys from nine African countries, the rule reaches about 1.8 times as many poor people as the plug-in rule and matches its poverty-gap reduction with 6.7 percent less spending.

OTHER WORKING PAPERS

When and How to Pilot: Design Rules for Two-Wave Experiments

A pilot before the main wave yields variance estimates that can guide how to split the main sample between treatment and control. Balanced assignment ignores this evidence. Feasible Neyman allocation uses it, but with a small pilot it overreacts to noise and can do worse than balance. I propose a Conditional Minimax Regret (CMR) rule that minimizes worst-case regret over a finite-sample confidence set for the two variances. CMR retains balance's worst-case protection with high probability, converges to the Neyman allocation as the pilot grows, and attains the minimax-regret rate. In simulations calibrated to four field experiments, CMR avoids feasible Neyman's severe small-pilot losses while matching its large-pilot gains.

Two-Way Effects Models: A Nonparametric Empirical Bayes Approach

with Cole Davis

Draft available upon request

We develop a nonparametric empirical Bayes framework for models with unit and cluster effects. The framework allows unit effects to depend on latent cluster effects, relaxing independence restrictions in shrinkage estimation.

PUBLICATIONS

Birds of a Feather Collude Together: Subnational Alignment and Corruption

with Leopoldo Fergusson, Arturo Harker, and Carlos Molina

Forthcoming in the American Political Science Review

Using close elections in Colombia, we study how partisan alignment between municipal mayors and departmental governors affects corruption. Alignment increases the fabrication of “ghost” students, which inflates education transfers to municipalities, by 0.3 standard deviations, without improving genuine enrollment or student performance. It also raises discretionary hiring, patronage-based outsourcing, electoral fraud risk, and cross-level appointments of politicians' relatives. Aligned politicians nonetheless enjoy better future electoral prospects.

SOFTWARE

cmrdesign (R and Python)

[Documentation](#) | [Code](#) | [CRAN](#) | [PyPI](#)

Uses pilot data to choose main-wave treatment and control sample shares and report a finite-sample regret certificate. Supports two-arm, multi-arm, and stratified experiments.

RESEARCH AND INDUSTRY EXPERIENCE

Brown University, Department of Economics

June 2023–Jan. 2026

Research Assistant to Soonwoo Kwon, Providence, RI

Conducted simulation analyses and empirical applications for research on shrinkage estimation of fixed effects, inference in regression discontinuity designs under monotonicity, and bias-aware inference in regularized regression models.

Amazon, Delivery Experience

May–Aug. 2025

Economist Intern, Bellevue, WA

Owned a research project evaluating a nationwide U.S. delivery-speed program. Formulated the causal question, acquired data using SQL, and implemented difference-in-differences and shift-share IV designs in Python to estimate direct and geographic spillover effects and quantify the program's total effect on customer delivery experience. Presented findings and recommendations to senior economists and program decision-makers.

Universidad de los Andes, Department of Economics

Aug. 2017–July 2021

Research Assistant to Leopoldo Fergusson and James A. Robinson, Bogotá, Colombia

Built and linked administrative, archival, and web-scraped datasets for research on conflict, corruption, and state capacity. Implemented instrumental-variables and regression-discontinuity designs, examined mechanisms and heterogeneous effects, and conducted robustness and replication checks. Contributed empirical analysis and writing to Acemoglu, Fergusson, and Johnson's study of population growth and conflict, published in the *Review of Economic Studies*.

SELECTED PRESENTATIONS

Midwest Econometrics Group Annual Meeting, Cincinnati 2026

Poverty Targeting with Imperfect Information (accepted for presentation)

World Congress of the Econometric Society, Seoul 2025

Poverty Targeting with Imperfect Information

Advances with Field Experiments, Chicago 2025

When and How to Pilot: Design Rules for Two-Wave Experiments

AWARDS AND FELLOWSHIPS

Merit Dissertation Fellowship, Department of Economics, Brown University 2026

International Travel Fund, Graduate School, Brown University 2025

Conference Travel Fund, Graduate School, Brown University 2025

TEACHING EXPERIENCE

Brown University, Teaching Assistant

Applied Econometrics II (Ph.D.), Peter Hull Spring 2026

Applied Econometrics I (Ph.D.), Toru Kitagawa Fall 2024

Using Big Data to Solve Economic and Social Problems (undergraduate), John N. Friedman Fall 2022 and 2023

Universidad de los Andes, Teaching Assistant Aug. 2018–June 2020

Advanced Econometrics (graduate), Raquel Bernal

Political Underpinnings of Prosperity and Poverty (graduate), James A. Robinson

Thinking Problems (undergraduate), Tomás Rodríguez

ACADEMIC SERVICE

Organizer, Brown Metrics Coffee 2024–26

TECHNICAL SKILLS AND LANGUAGES

Software: R, Python, Stata, MATLAB, SQL, Git, LaTeX

Languages: Spanish (native), English (fluent), French (fluent)